

1959.

Wow, that is very early. I remember being a senior in college 11 years later and trying to run an econometric model on the IBM 360. You would type the program on a stack of punch cards, and if there was one misplaced comma, the program would bomb, and you would have to submit the program all over again.

That was my experience as well. The IBM 704 was new and rare at the time, and it served 30 New England universities. As a faculty member, I was able to reserve some time on it. I did not know much about programming, but I ended up developing things that programmers already knew about, such as subroutines. I modularized the program and tested blocks separately to make sure they ran and got the results I thought they should get. Then I put the blocks together. I would take a block in for testing—the block might be something like calculating doubling down expectations. Similar to your experience, two and three days later, my cards would come back with a rubber band and piece of paper around them, indicating the grammatical mistakes I had made. Initially I was bogged down with grammatical errors, but then I got much more accurate in doing it, and things began to run much more smoothly.

By 1960, I started getting results back, and they were really exciting. When I ran the combination of four aces out, the edge went to -2.5 percent. That result implied that if there were four extra aces in the deck, the odds would go to $+2.5$ percent. You might ask, how can you have four extra aces? Of course, you can't, but if you are down to half a deck and none of the aces are out, the odds are the same as a full deck with eight aces.

Did you calculate the odds by varying one card at a time and holding everything else constant?

Yes, that tied into the image I had when I read the original paper back in UCLA. You can picture the blackjack probability problem as a 10-dimensional space, with the fraction of each card varying along a single axis. The fraction of each card is one-thirteenth, except for the cards with a 10 value, which pool and have a combined fraction of four-thirteenths. You can think of any deck—that is, any combination of cards—as being a